

a Task hierarchy

L1

Task Progress Memory

Rice cooking

Pot in sinkFirst rinseDrain waterSecond rinseCurrent state

L2

Self-induced Anomaly Memory

Steamed buns

Bun slips onto floorDiscard soiled bunWipe floor stainsAdd fresh bunPlate restored

L3

External Intervention Memory

Meal box

Two slices preparedBox closedPerson takes sliceOne slice remainsMasked box state

L4

Compound Long-Horizon Memory

Anomaly + intervention

Self-caused spillWipe up the spillOther person takes trayFruit falls from trayCleanup pending

b Paired evaluation

P

Next-action prediction

Goal: Heat the steamed buns

History + current observation

What is the most appropriate next action?

A

Put the buns back into the refrigerator.

B

Pick up the plate and approach the steamer.

C

Open the refrigerator and take another bun out.

D

Take another plate out of the kitchen cabinet.

C

Causal traceback

Given action: Pick up plate and approach steamer

Which historical event necessitated this action?

A

Take steamer lid off

B

Turn on kitchen faucet

D

Plate slips and bun falls onto the floor

C

Step towards dustbin

